**26900050** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

## REGULAR EXAMINATION October 2026

## Fifth SEMESTER

## Discipline Specific Core Course (DSC) - MG5DSCBTS300 - DEVELOPMENTAL BIOLOGY

(2024 ADMISSION ONWARDS)

**Duration: 1 Hour 30 Minutes** **Maximum Marks: 50**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Who proposed the Germ Plasm Theory? [K] / [CO1]

   a). August Weismann b). Aristotle

   c). Wilhelm Roux d). Hans Spemann

2. During normal cleavage, the chromosome number of blastomeres is: [K] / [CO2]

   a). Reduced by half b). Doubled

   c). Maintained d). Completely lost

3. The darker pigmented region of a frog egg is called the: [K] / [CO4]

   a). Vegetal hemisphere b). Blastopore

   c). Animal hemisphere d). Archenteron

4. An egg containing a large amount of yolk is called: [U] / [CO2]

   a). Microlecithal b). Alecithal

   c). Macrolecithal d). Isolecithal

5. The lens develops from the: [K] / [CO4]

a). Neural ectoderm b). Surface ectoderm

c). Mesoderm d). Endoderm

6. In epiboly, cells mainly: [U] / [CO2]

a). Move inward individually b). Spread over other cells

c). Divide meiotically d). Form gametes

7. Which of the following best describes the focus of chemical developmental biology? [K] / [CO1]

a). Ecosystems b). Chemical substances and molecular interactions

c). Climatic factors d). Environmental pollution

8. Which process produces specialized cell types during development? [K] / [CO5]

a). Cell differentiation b). Fertilization

c). Cleavage only d). Cytokinesis only

9. Salamanders show greater regenerative ability than adult mammals mainly because they can: [K] / [CO3]

a). Undergo repeated fertilization b). Produce more gametes

c). Re-form complex tissues after injury d). Increase chromosome number

10. A commonly used method for constructing fate maps involves: [U] / [CO2]

a). Cell marking b). Blood grouping

c). Chromatography d). Electrophoresis

**[1x10 = 10]**

## Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. Name the major phases of the menstrual cycle. [K] / [CO2]

12. A female insect produces offspring without mating. Identify the reproductive process and explain your answer. [A] / [CO2]

13. What is the source of alpha-fetoprotein during pregnancy? [K] / [CO3]

14. Describe neurulation and notogenesis in frog embryo. [K] / [CO4]

15. What is the apical ectodermal ridge (AER)? [K] / [CO3]

16. Name the main parts of a sperm. [K] / [CO2]

17. What is the zona pellucida? [U] / [CO2]

18. Mention any two biochemical changes occurring during frog metamorphosis. [K] / [CO4]

19. Name any two derivatives of endoderm. [K] / [CO2]

20. Define pregnancy. [K] / [CO2]

21. Define prenatal diagnosis. [K] / [CO3]

22. Name the three major regions into which the mesoderm differentiates. [K] / [CO4]

**[2x10 = 20]**

## Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Explain the role of segmentation genes in Drosophila development. [U] / [CO5]

24. Differentiate between gametogenesis and spermatogenesis. [K] / [CO2]

25. Explain the hormonal control of male reproductive function. [K] / [CO2]

26. Differentiate between the morula and blastula. [K] / [CO2]

27. Explain the role of seminal vesicles, prostate gland and Cowper's glands in the formation of semen. [K] / [CO2]

28. Describe the role of the dorsal lip of the blastopore as an organizer. [U] / [CO4]

29. Define embryo transfer technology and describe its basic steps. [U] / [CO4]

30. Explain the different stages of oogenesis. [U] / [CO2]

**[4x5 = 20]**